Computing since the 1940s

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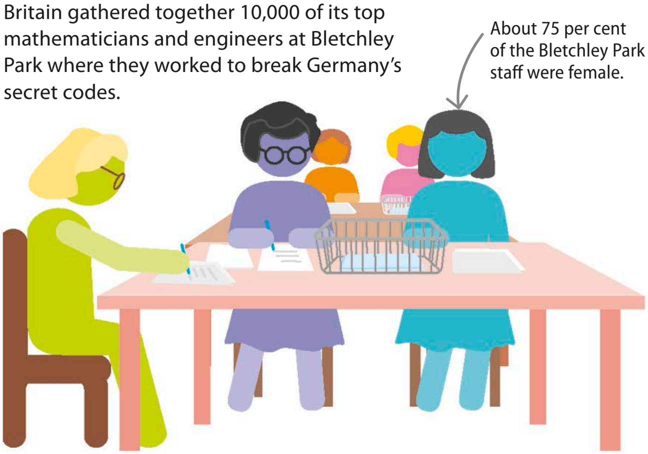
Computing has advanced dramatically since the 1940s. Starting as an abstract, mathematical pursuit carried out by academics, it's now a part of life for a huge number of people.

WWII and computers

World War II was a catalyst for the development of electronic computing. The German forces used a cryptography machine called Enigma to make their messages secret. English computer scientist Alan Turing led a team that deciphered Enigma by building a computer called Bombe. This computer contained hundreds of moving parts, and was a step away from being the forerunner of today's computers.

▽ **Code-breakers during WWII**
Britain gathered together 10,000 of its top mathematicians and engineers at Bletchley Park where they worked to break Germany's secret codes.



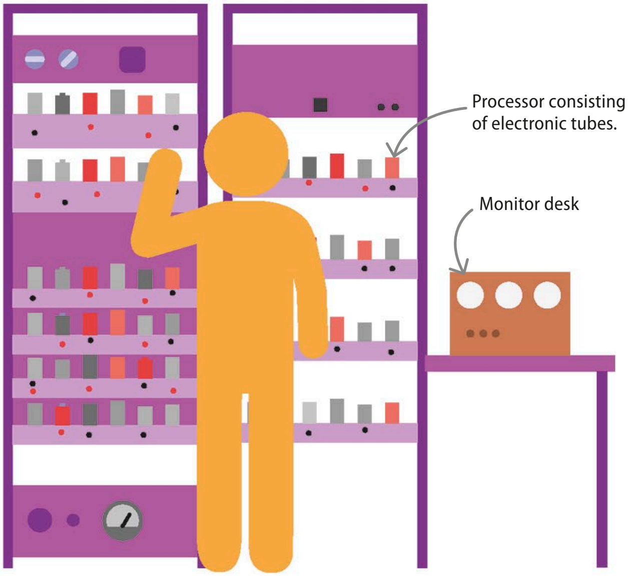
REAL WORLD

Human computers

The word "computer" used to refer to humans who calculated mathematical results using pencil and paper. From the late 19th century to the mid-20th century, human computers were often women, including American mathematician Katherine Johnson. Their work was essential in a number of fields, for example, computing data for early space flights at NASA.

Stored program computers

Colossus, built in 1943, was a computer that had a fixed function: to break coded messages. The same was true of the ENIAC (Electronic Numerical Integrator And Computer), developed around the same time, which calculated the paths of missiles for the US Army. Changing the program of either of these computers involved rewiring the machine and physically pulling switches. The first practical general-purpose stored-program electronic computer was EDSAC (electronic delay storage automatic calculator), which ran its first program in 1949. It could be reprogrammed with ease, and typically worked for 35 hours a week, carrying out calculations that a human would find complex and time-consuming.



△ **EDSAC**
The developers of the ENIAC went on to design and build the EDSAC, the first computer that stored programs and data in the same machine.